Lesson: Introduction to C

CSE 4107 : Structured Programming I

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Origins of C

- A byproduct of UNIX operating systems
- Developed at Bell Laboratories by Ken Thompson, Dennis Ritchie and others
- UNIX was written in Assembly language
- Debugging complications inspired Thompson to write a new language - B
- B was based on BCPL (a systems programming language)

Origins of C

- By 1971, Ritchie began to develop an extended version of B, which he named **NB** (“New B”)
- As the language diverged from B, he changed its name to C
- UNIX was rewritten in C in 1973

Chronological evolution of C language

Standardization of C

- *K&R C*
    - Described in Kernighan and Ritchie, *The C Programming Language (1978)*
    - Became the De facto standard
- Popularized in 1970s
- As popularity grew, different C compiler processed features differently in the 1980s

Standardization of C

- *C89/C90*
    - ANSI* standard approved in 1989
- *ANSI C/C99*
    - New standard in 1999 by ANSI
- Current Standard is : C11

* ANSI (American National Standards Institute)

C based languages

- ***C++*** includes all the features of C, but adds feature support for object oriented programming
- ***Java*** based on C++ and therefore inherits many features
- ***C#*** is more recent language derived from Java and C++
- ***Perl*** another scripting language, adopted many features of C
- ***JavaScript*** uses syntax influenced by C

Properties of C

- Low level language
  - Provides access to machine level concepts
- Comparatively small
  - C keeps itself limited by relying heavily on a library of standard functions
- Permissive
  - Provides more freedom compared to other higher level languages (which is also more risky)